

# SMART INDIA HACKATHON 2025



- Problem Statement ID – SIH25071
- Problem Statement Title-AI-Based Rockfall Prediction and Alert System for Open-Pit Mines
- Theme- Disaster Management
- PS Category- Software
- Team ID-
- Team Name - GeoGuards



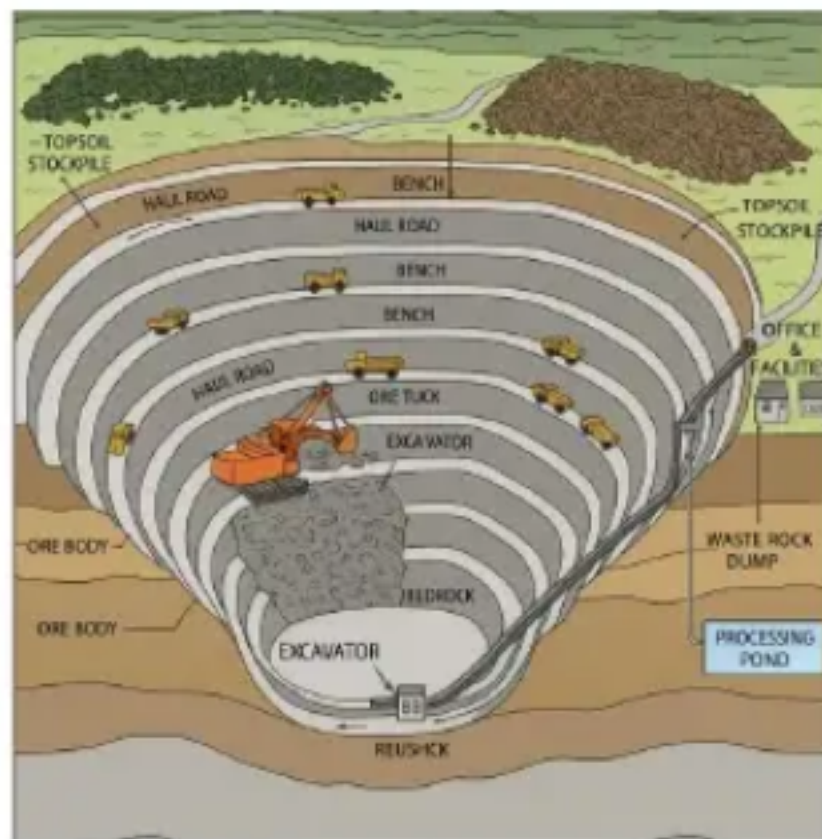
# IDEA TITLE

## • Proposed Solution :

Developing a web based AI/ML software to predict rockfall probability based on simulated weather datasets (rainfall, humidity, temperature, soil conditions etc) and Provide a dashboard with real-time alerts (SMS/Email).

## Addressing the Problem :

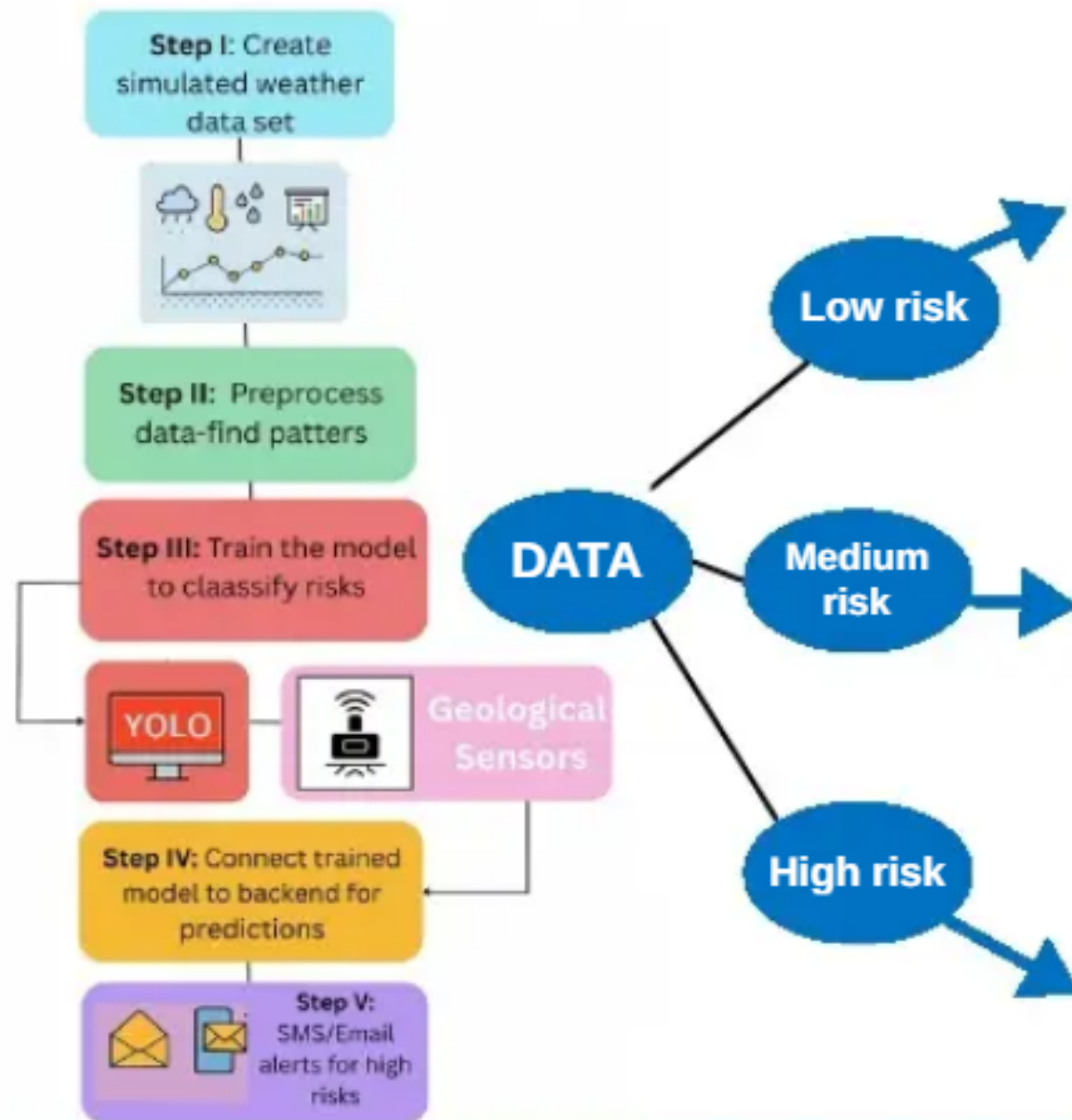
- By giving early warning of rockfalls in open pit mines, preventing lifelosses.
- Helps operators and workers to take early safety measures.



## Solution overview :

- Geological risk prediction via ml model.
- Integration of vibration sensors and models to learn patterns.
- Dashboard + alerts showing real-time risk levels.

## Technologies :



## Feasibility :

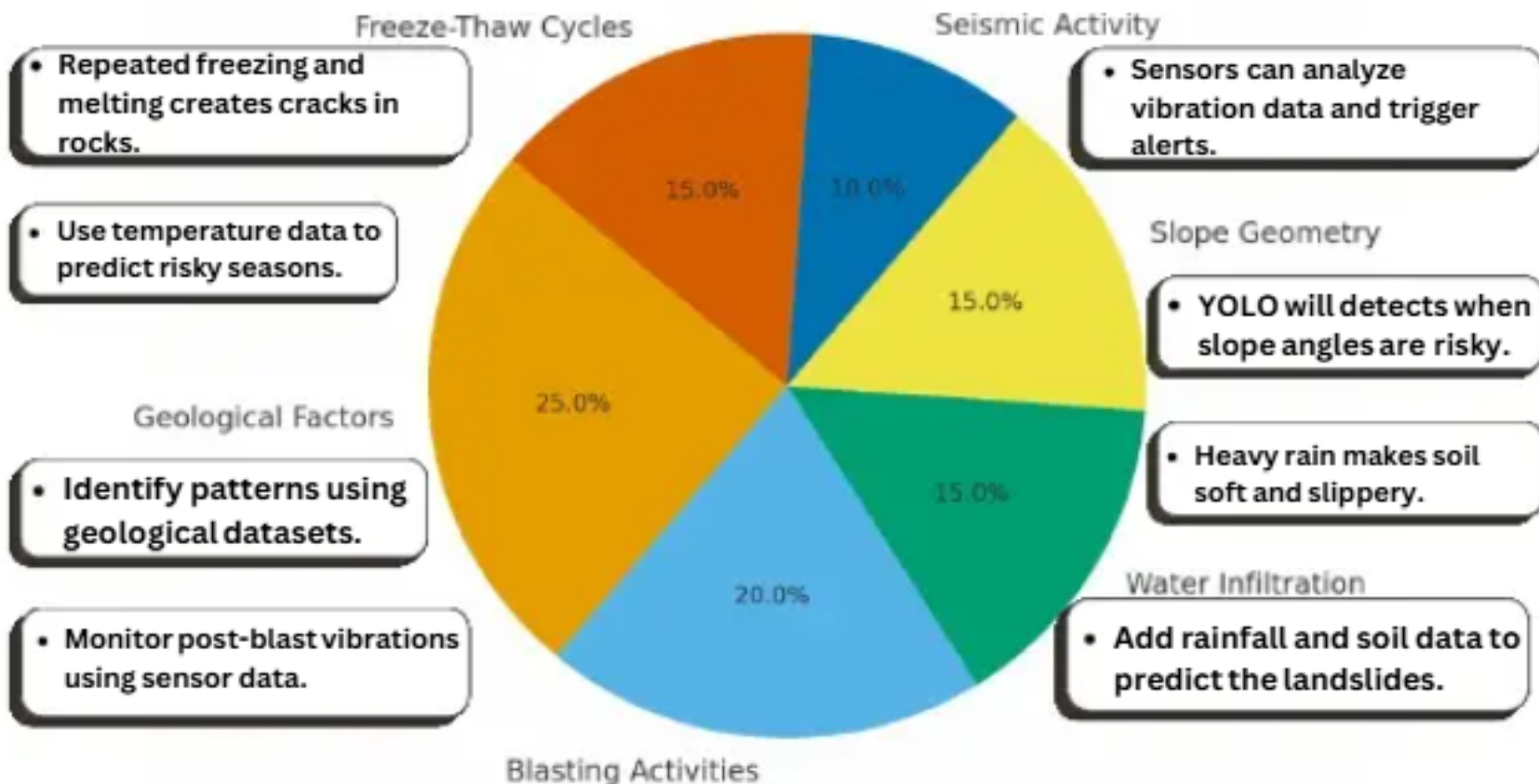
- **Time-feasible** – dataset → training → dashboard.

- **Scalable** – The system will be expanded with real IoT sensors.

- **Low cost** – Can be developed using open-source libraries

- **Modular design** – System components allow to be updated independently.

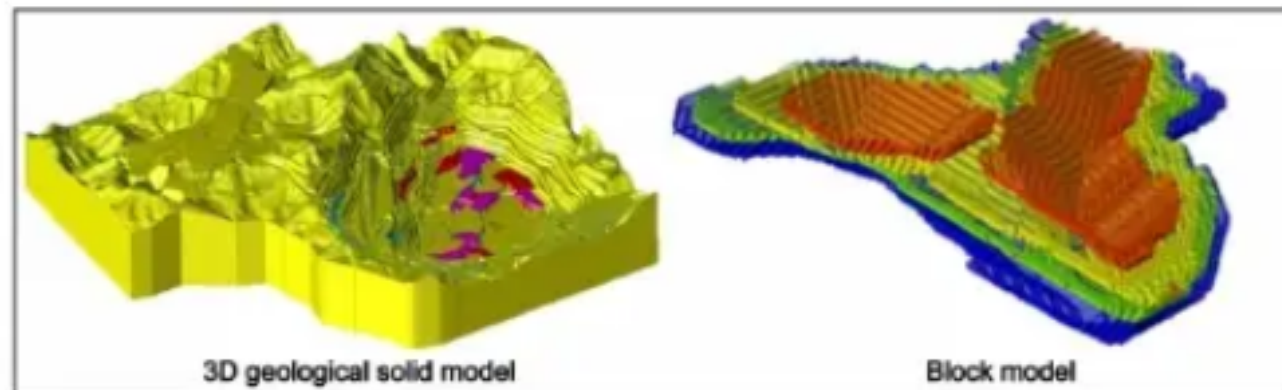
## Challenges in overcoming Rockfall causes:



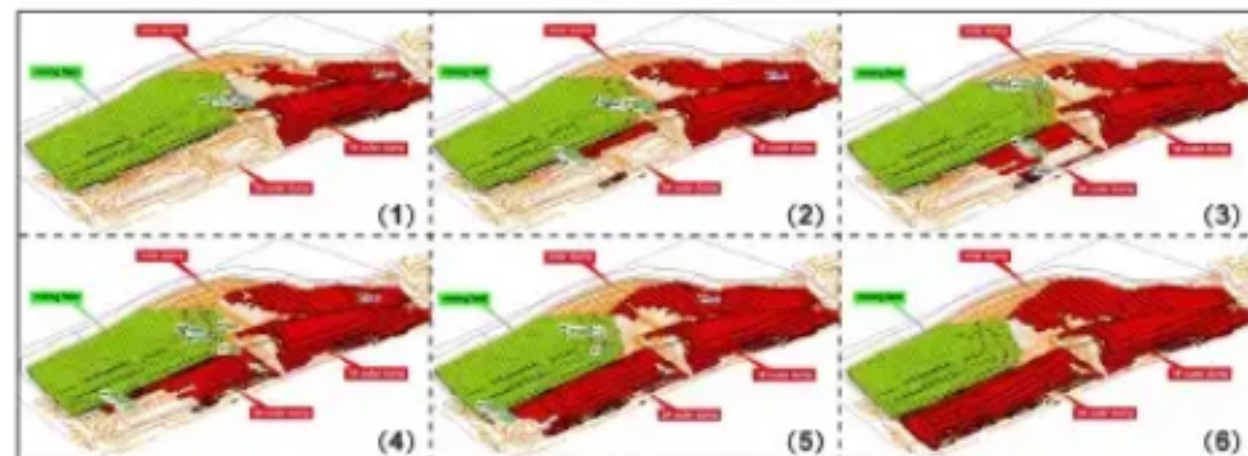


## Impacts :

- Improves worker safety with early alerts
- Helps operators make better decisions
- Cuts downtime and losses from rockfalls
- Builds trust with regulators and stakeholders
- Shows AI as a cost-effective safety solution



(a) open-pit mine with regular block model



(b) Dynamic evolution of open-pit mine with regular block model

### Social

- Saves workers' lives → AI alerts (SMS/Dashboard)
- Builds worker trust by real-time monitoring.

## Benefits :

### Economic

- Protects equipment through YOLO risk prediction.
- Budget friendly as it uses open-source AI and geological sensors.

### Environmental

- Stops slope failures
- → weather-based AI models
- Cuts resource waste
- Prevents uncontrolled slope failures, reducing land damage.



## A. Research

- Studies show that weather conditions (rainfall, wind, soil moisture) play a key role in triggering rockfalls.
- The YOLO algorithm, though widely used in object detection, is being adapted here in a unique way for predicting geological risks.
- <https://www.mdpi.com/1424-8220/25/10/3131>

## B. References

1. Open-source AI projects and datasets (Kaggle, GitHub mining datasets).
2. SIH Problem Statement documents & mining safety standards.